Liquidity ratios

INTRODUCTION TO FINANCIAL STATEMENTS IN POWER BI

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Performance measurements

Key Performance Indicators (KPIs) - set of measurements used to determine a company's performance.

- Determined by the firm
- Useful if well-defined

[1] https://www.investopedia.com/ask/answers/06/amortizationvsdepreciation.asp

Financial ratio analysis

The relationship among selected items of financial statement data.

- Liquidity ratios
- Profitability Ratios

[1] Weygandt, Jerry J., et al. Accounting Principles, John Wiley & Sons, 2015.

Liquidity ratios

- Short-term ability of the company to pay obligations
- **Example:**
  - Every month - pay rent, student loan, car loan
  - Liquidity - ability to pay these bills
- Three main liquidity ratios:
  - Current ratio
  - Acid-test
  - Inventory Turnover

Current ratio

- Compares current assets to current liabilities
- Measures short-term ability to pay debt

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

$$\frac{\$1{,}020{,}000}{\$344{,}500} = 2.96{:}1$$

Larry's Automotive.
Balance Sheet (partial)

	2017	2018
Assets		
Current assets	$1,020,000	$ 945,000
Plant assets (net)	800,000	632,500
Intangible assets	15,000	17,500
Total assets	$1,835,000	$1,595,000
Liabilities		
Current liabilities	$ 344,500	$ 303,000
Long-term liabilities	487,500	497,000
Total liabilities	832,000	800,000
Stockholders' Equity		
Common stock, $1 par	275,400	270,000
Retained earnings	727,600	525,000
Total stockholders' equity	1,003,000	795,000
Total liabilities and stockholders' equity	$1,835,000	$1,595,000

[1] Weygandt, Jerry J., et al. Accounting Principles, John Wiley & Sons, 2015.

Acid-test ratio

- Measures short-term liquidity

$$\text{Acid-Test Ratio} = \frac{\text{Cash + Short-Term Investments + Accounts Receivable (Net)}}{\text{Current Liabilities}}$$

$$\frac{\$100{,}000 + \$20{,}000 + \$230{,}000}{\$344{,}500} = 1.02{:}1$$

Larry's Automotive.
Balance Sheet (partial)

	2013
Current assets	
Cash	**$ 100,000**
Short-term investments	**20,000**
Accounts receivable (net*)	**230,000**
Inventory	620,000
Prepaid expenses	50,000
Total current assets	$1,020,000

*Allowance for doubtful accounts is $10,000 at the end of eac

[1] Weygandt, Jerry J., et al. Accounting Principles, John Wiley & Sons, 2015.

Inventory turnover

- Measures the number of times the inventory is sold during a certain period.

$$\text{Inventory Turnover} = \frac{\text{Cost of Goods Sold}}{\text{Average Inventory}}$$

$$\frac{\$1{,}281{,}000}{\left[\frac{\$500{,}000 + \$620{,}000}{2}\right]} = 2.3 \text{ times}$$

- Days in Inventory

```
365 / 2.3 = Every 159 days
```

Larry's Automotive
Balance Sheet (partial)

	2017	2018
Current assets		
Cash	**$ 100,000**	**$155,000**
Short-term investments	**20,000**	**70,000**
Accounts receivable (net*)	**230,000**	**180,000**
Inventory	620,000	500,000
Prepaid expenses	50,000	40,000
Total current assets	$1,020,000	$ 945,000

Larry's Automotive
Condensed Income Statements
For the Years Ended December 31

	2017
Sales revenue	$2,195,000
Sales returns and allowances	98,000
Net sales	2,097,000
Cost of goods sold	1,281,000
Gross profit	816,000
Selling expenses	253,000
Administrative expenses	104,000

[1] Weygandt, Jerry J., et al. Accounting Principles, John Wiley & Sons, 2015.

Let's practice!

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Liquidity dashboard

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Profitability analysis

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Profitability ratios

- Measures income success over time
- Ratios Include:
  - Profit Margin
  - Asset Turnover
  - Return on Assets
- Indicate a company's income-earning capacity and affect
  - Company's growth
  - Ability to take out loans
  - Attract investors

[1] Weygandt, Jerry J., et al. Accounting Principles, John Wiley & Sons, 2015.

Profit margin

- Percentage of each dollar of sales that results in net income.

$$\text{Profit Margin} = \frac{\text{Net Income}}{\text{Net Sales}}$$

$$\frac{\$263{,}800}{\$2{,}097{,}000} = 12.6\%$$

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Gross profit	816,000
Selling expenses	253,000
Administrative expenses	104,000
Total operating expenses	357,000
Income from operations	459,000
Other revenues and gains	
Interest and dividends	9,000
Other expenses and losses	
Interest expense	36,000
Income before income taxes	432,000
Income tax expense	168,200
Net income	$ 263,800

[1] Weygandt, Jerry J., et al. Accounting Principles, John Wiley & Sons, 2015.

Asset turnover

- Efficiency in assets used to generate sales.
- **Example - Cookie business:**
  - Several ovens cook hundreds of cookies

$$\text{Asset Turnover} = \frac{\text{Net Sales}}{\text{Average Total Assets}}$$

$$\frac{\$2{,}097{,}000}{\left[\frac{\$1{,}595{,}000 + \$1{,}835{,}000}{2}\right]} = 1.2 \text{ times}$$

Larry's Automotive.
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Return on assets (ROA)

- Measures overall profitability in relation to Total Assets

$$\text{Return on Assets} = \frac{\text{Net Income}}{\text{Average Total Assets}}$$

$$\frac{\$263{,}800}{\left[\frac{\$1{,}595{,}000 + \$1{,}835{,}000}{2}\right]} = 15.4\,\%$$

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Larry's Automotive
Condensed Balance Sheets

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Profit dashboard

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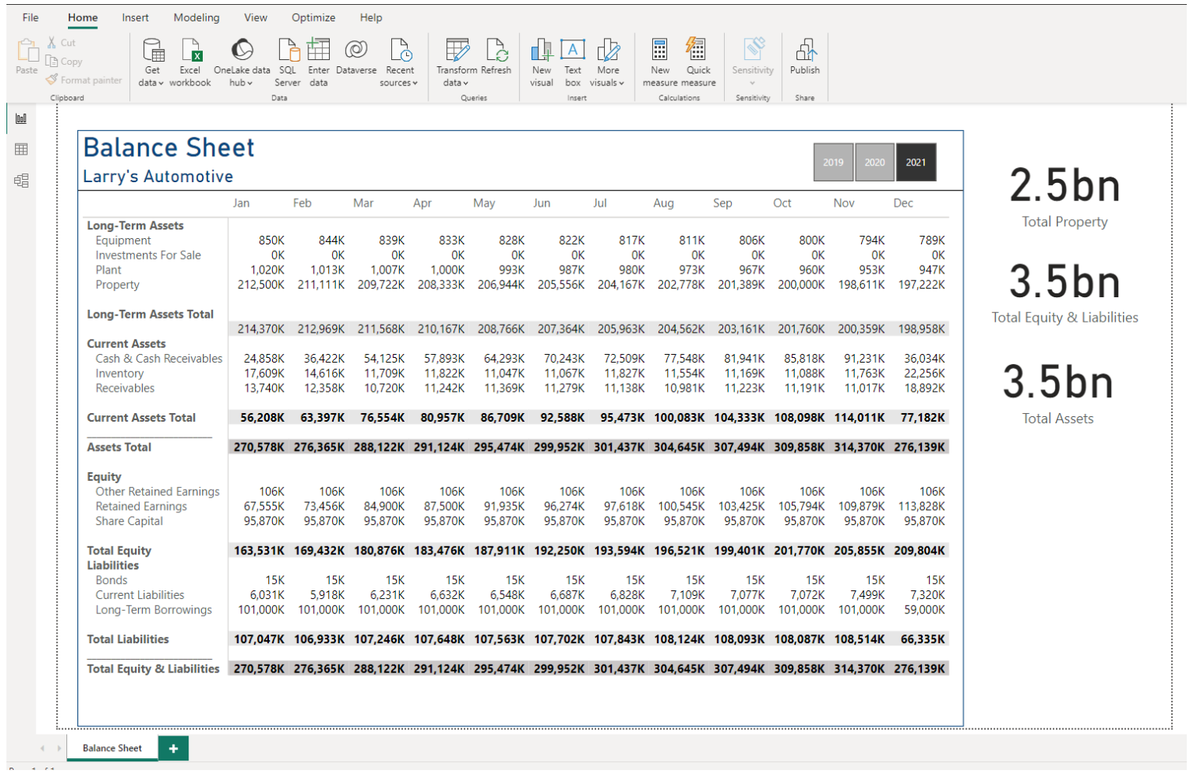
The accounting equation

- Balance Sheet
- Income Statement
- Accounting Equation
- Detection of Trends
- Correction of Errors



Balance Sheet

Larry's Automotive

2019 | 2020 | 2021

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Long-Term Assets												
Equipment	850K	844K	839K	833K	828K	822K	817K	811K	806K	800K	794K	789K
Investments For Sale	0K	0K	0K	0K	0K	0K	0K	0K	0K	0K	0K	0K
Plant	1,020K	1,013K	1,007K	1,000K	993K	987K	980K	973K	967K	960K	953K	947K
Property	212,500K	211,111K	209,722K	208,333K	206,944K	205,556K	204,167K	202,778K	201,389K	200,000K	198,611K	197,222K
Long-Term Assets Total	214,370K	212,969K	211,568K	210,167K	208,766K	207,364K	205,963K	204,562K	203,161K	201,760K	200,359K	198,958K
Current Assets												
Cash & Cash Receivables	24,858K	36,422K	54,125K	57,893K	64,293K	70,243K	72,509K	77,548K	81,941K	85,818K	91,231K	36,034K
Inventory	17,609K	14,616K	11,709K	11,822K	11,047K	11,067K	11,827K	11,554K	11,169K	11,088K	11,763K	22,256K
Receivables	13,740K	12,358K	10,720K	11,242K	11,369K	11,279K	11,138K	10,981K	11,223K	11,191K	11,017K	18,892K
Current Assets Total	**56,208K**	**63,397K**	**76,554K**	**80,957K**	**86,709K**	**92,588K**	**95,473K**	**100,083K**	**104,333K**	**108,098K**	**114,011K**	**77,182K**
Assets Total	**270,578K**	**276,365K**	**288,122K**	**291,124K**	**295,474K**	**299,952K**	**301,437K**	**304,645K**	**307,494K**	**309,858K**	**314,370K**	**276,139K**
Equity												
Other Retained Earnings	106K	106K	106K	106K	106K	106K	106K	106K	106K	106K	106K	106K
Retained Earnings	67,555K	73,456K	84,900K	87,500K	91,935K	96,274K	97,618K	100,545K	103,425K	105,794K	109,879K	113,828K
Share Capital	95,870K	95,870K	95,870K	95,870K	95,870K	95,870K	95,870K	95,870K	95,870K	95,870K	95,870K	95,870K
Total Equity	**163,531K**	**169,432K**	**180,876K**	**183,476K**	**187,911K**	**192,250K**	**193,594K**	**196,521K**	**199,401K**	**201,770K**	**205,855K**	**209,804K**
Liabilities												
Bonds	15K	15K	15K	15K	15K	15K	15K	15K	15K	15K	15K	15K
Current Liabilities	6,031K	5,918K	6,231K	6,632K	6,548K	6,687K	6,828K	7,109K	7,077K	7,072K	7,499K	7,320K
Long-Term Borrowings	101,000K	101,000K	101,000K	101,000K	101,000K	101,000K	101,000K	101,000K	101,000K	101,000K	101,000K	59,000K
Total Liabilities	**107,047K**	**106,933K**	**107,246K**	**107,648K**	**107,563K**	**107,702K**	**107,843K**	**108,124K**	**108,093K**	**108,087K**	**108,514K**	**66,335K**
Total Equity & Liabilities	**270,578K**	**276,365K**	**288,122K**	**291,124K**	**295,474K**	**299,952K**	**301,437K**	**304,645K**	**307,494K**	**309,858K**	**314,370K**	**276,139K**

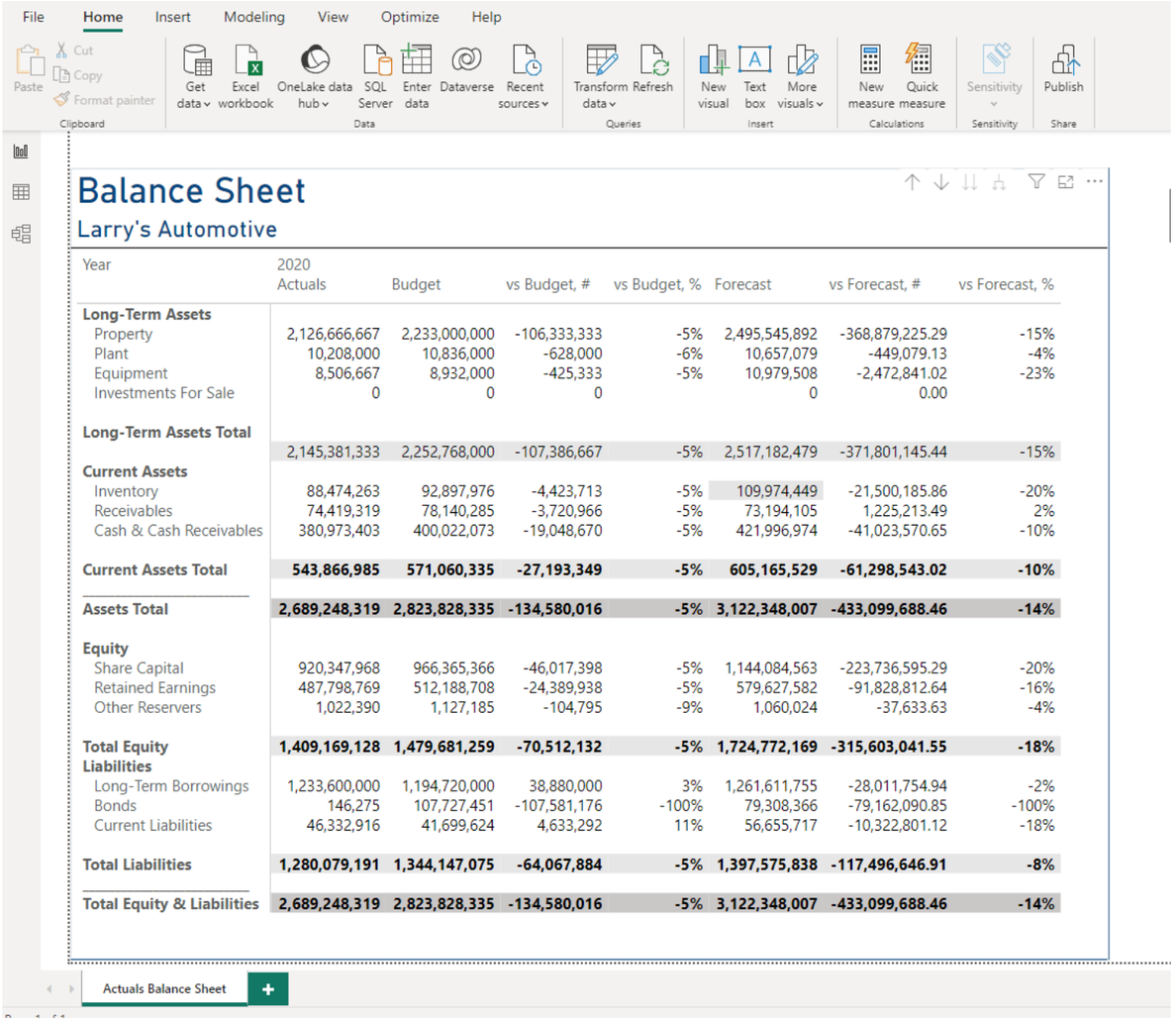
2.5bn
Total Property

3.5bn
Total Equity & Liabilities

3.5bn
Total Assets

Budgetary planning

- Limit financial surprises through budgeting and forecasting
- Created financial statements
  - included budgeted and forecasted numbers
- Calculated variance
- Analyzed actual vs. budget and forecast



Balance Sheet

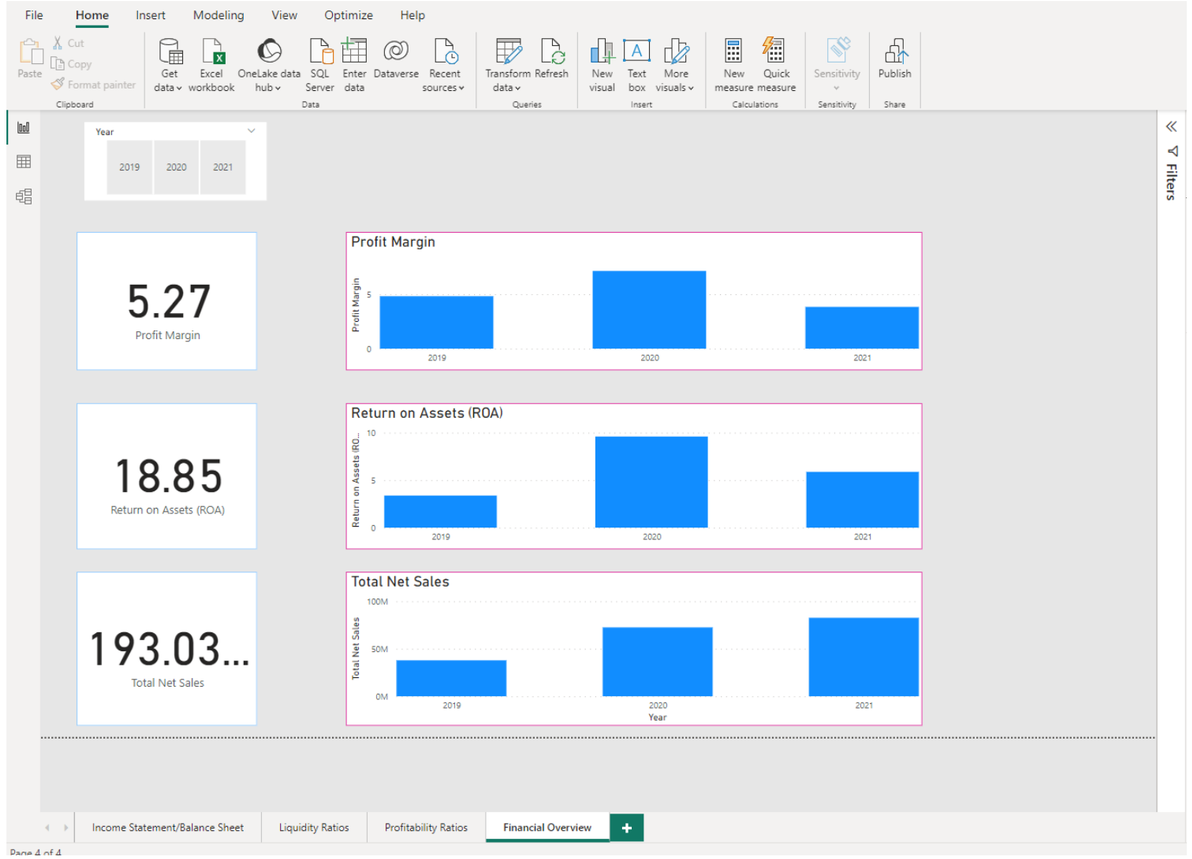
Larry's Automotive

Year	2020 Actuals	Budget	vs Budget, #	vs Budget, %	Forecast	vs Forecast, #	vs Forecast, %
Long-Term Assets							
Property	2,126,666,667	2,233,000,000	-106,333,333	-5%	2,495,545,892	-368,879,225.29	-15%
Plant	10,208,000	10,836,000	-628,000	-6%	10,657,079	-449,079.13	-4%
Equipment	8,506,667	8,932,000	-425,333	-5%	10,979,508	-2,472,841.02	-23%
Investments For Sale	0	0	0		0	0.00	
Long-Term Assets Total	2,145,381,333	2,252,768,000	-107,386,667	-5%	2,517,182,479	-371,801,145.44	-15%
Current Assets							
Inventory	88,474,263	92,897,976	-4,423,713	-5%	109,974,449	-21,500,185.86	-20%
Receivables	74,419,319	78,140,285	-3,720,966	-5%	73,194,105	1,225,213.49	2%
Cash & Cash Receivables	380,973,403	400,022,073	-19,048,670	-5%	421,996,974	-41,023,570.65	-10%
Current Assets Total	**543,866,985**	**571,060,335**	**-27,193,349**	**-5%**	**605,165,529**	**-61,298,543.02**	**-10%**
Assets Total	**2,689,248,319**	**2,823,828,335**	**-134,580,016**	**-5%**	**3,122,348,007**	**-433,099,688.46**	**-14%**
Equity							
Share Capital	920,347,968	966,365,366	-46,017,398	-5%	1,144,084,563	-223,736,595.29	-20%
Retained Earnings	487,798,769	512,188,708	-24,389,938	-5%	579,627,582	-91,828,812.64	-16%
Other Reservers	1,022,390	1,127,185	-104,795	-9%	1,060,024	-37,633.63	-4%
Total Equity	**1,409,169,128**	**1,479,681,259**	**-70,512,132**	**-5%**	**1,724,772,169**	**-315,603,041.55**	**-18%**
Liabilities							
Long-Term Borrowings	1,233,600,000	1,194,720,000	38,880,000	3%	1,261,611,755	-28,011,754.94	-2%
Bonds	146,275	107,727,451	-107,581,176	-100%	79,308,366	-79,162,090.85	-100%
Current Liabilities	46,332,916	41,699,624	4,633,292	11%	56,655,717	-10,322,801.12	-18%
Total Liabilities	**1,280,079,191**	**1,344,147,075**	**-64,067,884**	**-5%**	**1,397,575,838**	**-117,496,646.91**	**-8%**
Total Equity & Liabilities	**2,689,248,319**	**2,823,828,335**	**-134,580,016**	**-5%**	**3,122,348,007**	**-433,099,688.46**	**-14%**

Company performance and financial analysis

The relationship among selected items of financial statement data.

- Liquidity ratios
- Profitability Ratios



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